Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Joe Meirinho and Son Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
3354 W Whitmore Ave	Modesto	Stanislaus	95358

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1980

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X017-X044-X016-XXXX

B. OPERATORS

Meirinho, Steven

Operator name: Meirinho, Steven

Telephone no.: (209) 538-8284 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
3354 Whitmore Ave	Modesto	CA	95358

C. OWNERS

Meirinho, Joe

Legal owner name: Meirinho, Joe

Telephone no.: (209) 538-8284 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
3354 Whitmore Ave	Modesto	CA	95358

This owner is responsible for paying permit fees.

Meirinho, Pauline

Legal owner name: Meirinho, Pauline

Telephone no.: (209) 538-8284 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
3354 Whitmore Ave	Modesto	CA	95358

Meirinho, Steven

Legal owner name:	Meirinho, Steven	Telephone no.:	(209) 538-8284	
			Landline	Cellular
3354 Whitmore Ave	Modesto		CA	95358
Mailing Address Number and Street	City		State	Zip Code

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	81	0	0	125	0
Number under roof	984	0	0	0	0	0
Maximum number	989	92	0	0	136	0
Average number	984	81	0	0	125	0
Avg live weight (lbs)	1,400	1,450	0	0		

Predominant milk cow breed: Holstein

Average milk production: 75 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 27,367.37 *tons per reporting period*

Total nitrogen from manure: 357,755.55 *lbs per reporting period*

After ammonia losses (30% loss applied): 250,428.89 *lbs per reporting period*

Total phosphorus from manure: 59,489.19 *lbs per reporting period*

Total potassium from manure: 190,898.34 *lbs per reporting period*

Total salt from manure: 481,942.35 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 13,416,412 *gallons*

Total nitrogen generated: 38,466.45 *lbs*

Total phosphorus generated: 8,239.01 *lbs*

Total potassium generated: 75,122.38 *lbs*

Total salt generated: 528,339.45 *lbs*

	13,416,412	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	13,416,412	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
1	Ground water
2	Ground water
IW1	Ground water
IW2	Ground water
Patterson Irrigation District	Surface water

Source Description	Type
Turlock Irrigation District	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
02/25/2023	Corral solids	640.00 *ton*	Dry-weight	16.7		26,900.00	9,000.00	23,800.00		0.00

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	28,681.86	9,596.16	25,376.51	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	28,681.86	9,596.16	25,376.51	0.00

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
East Walker	67	67	2	manure	X017-X051-X006-XXXX X017-X051-X012-XXXX
HR North	16	16	2	manure	X056-X059-X018-XXXX
NRB	32	32	2	process wastewater	X017-X044-X015-XXXX
SM	24	24	1	none	X048-X004-X030-XXXX
SRB	33	33	2	process wastewater	X017-X044-X015-XXXX X017-X044-X016-XXXX
TD	12	12	2	process wastewater	X017-X044-X016-XXXX
West Walker	73	73	2	manure	X017-X051-X006-XXXX X017-X051-X012-XXXX
Totals for areas that were used for application	233	233	12		
Totals for areas that were not used for application	24	24	1		
Land application area totals	257	257	13		

B. CROPS AND HARVESTS

East Walker

Field name: East Walker

11/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 67 Plant date: 11/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	939.00 *ton*	Dry-weight		72.6	16,400.00	3,400.00	28,000.00		9.44

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	1,407.00
Total actual harvest content	14.01	125.95	26.11	215.05	725.01

East Walker

06/06/2023: Corn, silage

Crop: Corn, silage Acres planted: 67 Plant date: 06/06/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/28/2023	1,969.00 *ton*	Dry-weight		68.9	12,900.00	2,700.00	7,600.00		4.12

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	1,500.00
Total actual harvest content	29.39	235.80	49.35	138.92	753.11

HR North

Field name: HR North

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 16 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	222.00 *ton*	Dry-weight		71.2	18,100.00	3,200.00	27,600.00		9.36

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	1,407.00
Total actual harvest content	13.88	144.66	25.57	220.58	748.05

06/18/2023: Corn, silage

Crop: Corn, silage Acres planted: 16 Plant date: 06/18/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/28/2023	412.00 *ton*	Dry-weight		65.5	12,400.00	2,800.00	11,200.00		4.94

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	1,500.00
Total actual harvest content	25.75	220.32	49.75	199.00	877.71

NRB

Field name: NRB

11/12/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 32 Plant date: 11/12/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	498.00 *ton*	Dry-weight		74.9	22,900.00	4,300.00	40,300.00		13.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	1,494.00
Total actual harvest content	15.56	178.90	33.59	314.84	1,054.67

06/10/2023: Corn, silage

Crop: Corn, silage Acres planted: 32 Plant date: 06/10/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/18/2023	880.00 *ton*	Dry-weight		69.5	13,800.00	3,000.00	10,900.00		4.79

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	1,500.00
Total actual harvest content	27.50	231.50	50.33	182.85	803.52

SM

Field name: SM

10/01/2018: Almond, in shell

Crop: Almond, in shell Acres planted: 24 Plant date: 10/01/2018

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/01/2023	25.00 *ton*	Dry-weight		6.9	20,700.00	2,200.00	20,800.00		10.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	1.00	130.00	20.00	100.00	100.00
Total actual harvest content	1.04	40.15	4.27	40.34	211.41

SRB

Field name: SRB

11/12/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 33 Plant date: 11/12/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	466.00 *ton*	Dry-weight		74.3	23,500.00	4,200.00	39,500.00		12.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	1,494.00
Total actual harvest content	14.12	170.57	30.48	286.70	878.25

06/19/2023: Corn, silage

Crop: Corn, silage Acres planted: 33 Plant date: 06/19/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/18/2023	910.00 *ton*	Dry-weight		71.9	14,400.00	2,700.00	15,500.00		6.13

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	48.00	198.00	1,500.00
Total actual harvest content	27.58	223.17	41.84	240.21	950.00

TD

Field name: TD

11/05/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 12 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	164.00 *ton*	Dry-weight		76.8	26,900.00	4,300.00	49,700.00		14.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	1,494.00
Total actual harvest content	13.67	170.58	27.27	315.16	894.13

TD

06/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 12 Plant date: 06/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/18/2023	289.00 *ton*	Dry-weight		69.6	15,200.00	2,800.00	14,400.00		5.89

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	1,500.00
Total actual harvest content	24.08	222.57	41.00	210.85	862.45

West Walker

Field name: West Walker

11/06/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 73 Plant date: 11/06/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	1,119.00 *ton*	Dry-weight		79.8	22,300.00	5,000.00	44,100.00		9.63

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	1,407.00
Total actual harvest content	15.33	138.10	30.96	273.10	596.37

06/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 73 Plant date: 06/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/28/2023	2,069.00 *ton*	Dry-weight		68.9	12,600.00	3,100.00	8,800.00		4.27

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	1,500.00
Total actual harvest content	28.34	222.13	54.65	155.14	752.76

NUTRIENT BUDGET

A. LAND APPLICATIONS

East Walker - 11/08/2022: Oats, silage-soft dough

Field name: East Walker

Crop: Oats, silage-soft dough — Plant date: 11/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Dry Manure	Corral solids	139.36	41.55	243.24	0.00	540.00 *ton*
Application event totals		139.36	41.55	243.24	0.00	

10/07/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	5,567,378.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

East Walker - 06/06/2023: Corn, silage

Field name: East Walker

Crop: Corn, silage — Plant date: 06/06/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Dry Manure	Corral solids	242.35	96.63	146.50	0.00	559.00 *ton*
Application event totals		242.35	96.63	146.50	0.00	

05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	10,882,650.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

East Walker - 06/06/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	11,397,435.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN-32	Liquid commercial fertilizer	50.00	0.00	0.00	0.00	
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	11,097,431.00 *gal*
Application event totals		50.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	11,288,400.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	10,744,400.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	9,859,882.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

HR North - 11/03/2022: Oats, silage-soft dough

Field name: HR North

Crop: Oats, silage-soft dough

Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

HR North - 11/03/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/04/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Dry Manure	Corral solids	183.72	54.77	320.66	0.00	170.00 *ton*
Application event totals		183.72	54.77	320.66	0.00	

10/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	0.00	0.00	0.00	31.39	1,504,840.00 *gal*
Application event totals		0.00	0.00	0.00	31.39	

HR North - 06/18/2023: Corn, silage

Field name: HR North

Crop: Corn, silage

Plant date: 06/18/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Dry Manure	Corral solids	254.16	101.34	153.64	0.00	140.00 *ton*
Application event totals		254.16	101.34	153.64	0.00	

06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,384,850.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,584,400.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

HR North - 06/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,484,540.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,606,784.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	0.00	0.00	0.00	53.99	2,587,800.00 *gal*
Application event totals		0.00	0.00	0.00	53.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,456,995.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

NRB - 11/12/2022: Wheat, silage, soft dough

Field name: NRB

Crop: Wheat, silage, soft dough

Plant date: 11/12/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	97.12	15.86	127.30	1,146.81	2,104,110.00 *gal*
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,213,568.00 *gal*
Application event totals		97.12	15.86	127.30	1,146.81	

NRB - 11/12/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	115.03	13.50	178.12	978.99	521,400.00 *gal*
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,298,520.00 *gal*
Application event totals		115.03	13.50	178.12	978.99	

NRB - 06/10/2023: Corn, silage

Field name: NRB

Crop: Corn, silage

Plant date: 06/10/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	115.03	13.50	178.12	978.99	521,400.00 *gal*
Turlock Irrigation District	Surface water	0.00	0.00	0.00	53.04	5,084,400.00 *gal*
Application event totals		115.03	13.50	178.12	1,032.03	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	0.00	0.00	0.00	53.16	5,096,580.00 *gal*
Application event totals		0.00	0.00	0.00	53.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	5,102,331.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

NRB - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	72.80	21.41	179.66	1,288.84	782,000.00 *gal*
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	5,023,117.00 *gal*
Application event totals		72.80	21.41	179.66	1,288.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	72.80	21.41	179.66	1,288.84	782,000.00 *gal*
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	4,987,449.00 *gal*
Application event totals		72.80	21.41	179.66	1,288.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	40.44	11.89	99.80	715.95	434,400.00 *gal*
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	5,014,445.00 *gal*
Application event totals		40.44	11.89	99.80	715.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,125,668.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

SM - 10/01/2018: Almond, in shell

Field name: SM

Crop: Almond, in shell

Plant date: 10/01/2018

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

SM - 10/01/2018: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/05/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	1,855,088.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,406,784.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,506,784.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/25/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,506,784.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Liquid commercial fertilizer	30.00	0.00	0.00	0.00	
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,506,784.00 *gal*
Application event totals		30.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,506,784.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

SM - 10/01/2018: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,506,784.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	1,855,088.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,055,088.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

SRB - 11/12/2022: Wheat, silage, soft dough

Field name: SRB

Crop: Wheat, silage, soft dough

Plant date: 11/12/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	87.39	14.27	114.55	1,031.94	1,952,520.00 *gal*
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,586,580.00 *gal*
Application event totals		87.39	14.27	114.55	1,031.94	

SRB - 11/12/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	124.61	14.63	192.95	1,060.49	582,453.00 *gal*
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	4,589,660.00 *gal*
Application event totals		124.61	14.63	192.95	1,060.49	

SRB - 06/19/2023: Corn, silage

Field name: SRB

Crop: Corn, silage

Plant date: 06/19/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	5,598,600.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	95.45	28.07	235.55	1,689.77	1,057,300.00 *gal*
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	5,086,550.00 *gal*
Application event totals		95.45	28.07	235.55	1,689.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	5,123,445.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

SRB - 06/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	72.81	21.41	179.67	1,288.90	806,473.00 *gal*
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	5,023,554.00 *gal*
Application event totals		72.81	21.41	179.67	1,288.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	72.81	21.41	179.68	1,288.94	806,500.00 *gal*
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	4,987,225.00 *gal*
Application event totals		72.81	21.41	179.68	1,288.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	56.63	16.66	139.75	1,002.55	627,300.00 *gal*
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	4,996,885.00 *gal*
Application event totals		56.63	16.66	139.75	1,002.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	2,524,669.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

TD - 11/05/2022: Wheat, silage, soft dough

Field name: TD

Crop: Wheat, silage, soft dough

Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

TD - 11/05/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	84.94	13.87	111.33	1,002.95	690,060.00 *gal*
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	1,435,820.00 *gal*
Application event totals		84.94	13.87	111.33	1,002.95	

02/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	82.39	13.45	107.99	972.85	669,350.00 *gal*
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	1,229,240.00 *gal*
Application event totals		82.39	13.45	107.99	972.85	

TD - 06/17/2023: Corn, silage

Field name: TD

Crop: Corn, silage

Plant date: 06/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	111.38	32.76	274.87	1,971.83	448,650.00 *gal*
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	1,418,450.00 *gal*
Application event totals		111.38	32.76	274.87	1,971.83	

06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	0.00	0.00	0.00	51.60	1,855,088.00 *gal*
Application event totals		0.00	0.00	0.00	51.60	

TD - 06/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	55.08	16.20	135.92	975.02	221,845.00 *gal*
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	1,696,850.00 *gal*
Application event totals		55.08	16.20	135.92	975.02	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	1,532,520.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	101.45	29.84	250.36	1,796.03	408,651.00 *gal*
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	1,516,650.00 *gal*
Application event totals		101.45	29.84	250.36	1,796.03	

08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	1,496,800.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	1,529,240.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

West Walker - 11/06/2022: Oats, silage-soft dough

Field name: West Walker

Crop: Oats, silage-soft dough Plant date: 11/06/2022

West Walker - 11/06/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/15/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Dry Manure	Corral solids	161.07	48.02	281.12	0.00	680.00 *ton*
Application event totals		161.07	48.02	281.12	0.00	

10/22/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	4,414,800.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

West Walker - 06/17/2023: Corn, silage

Field name: West Walker

Crop: Corn, silage

Plant date: 06/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Dry Manure	Corral solids	228.80	91.22	138.31	0.00	575.00 *ton*
Application event totals		228.80	91.22	138.31	0.00	

06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	13,658,420.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	13,688,400.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

West Walker - 06/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN-32	Liquid commercial fertilizer	50.00	0.00	0.00	0.00	
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	13,893,452.00 *gal*
Application event totals		50.00	0.00	0.00	0.00	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	12,548,400.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	13,254,800.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

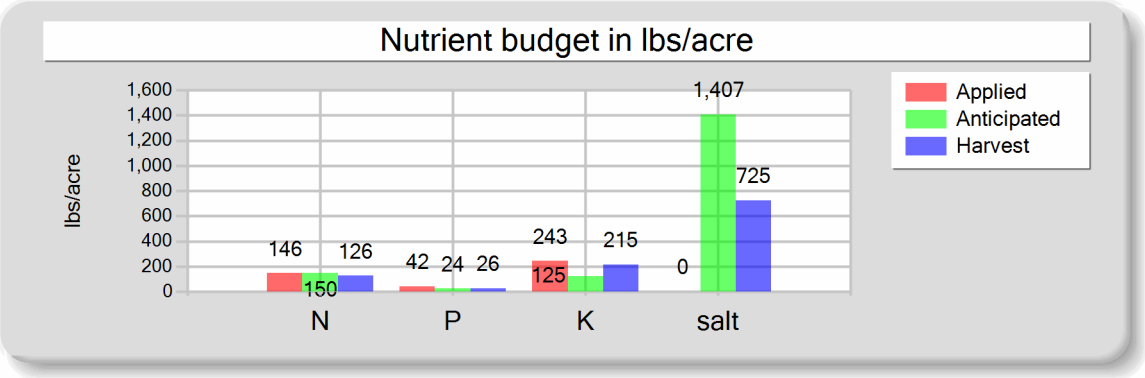
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Turlock Irrigation District	Surface water	0.00	0.00	0.00	0.00	1,556,995.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

B. NUTRIENT BUDGET

East Walker - 11/08/2022: Oats, silage-soft dough

Field name: East Walker | Crop: Oats, silage-soft dough | Plant date: 11/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	139.36	41.55	243.24	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	146.36	41.55	243.24	0.00
Anticipated crop nutrient removal	150.00	24.00	124.50	1,407.00
Actual crop nutrient removal	125.95	26.11	215.05	725.01
Nutrient balance	20.41	15.44	28.19	-725.01
Applied to removed ratio	1.16	1.59	1.13	0.00

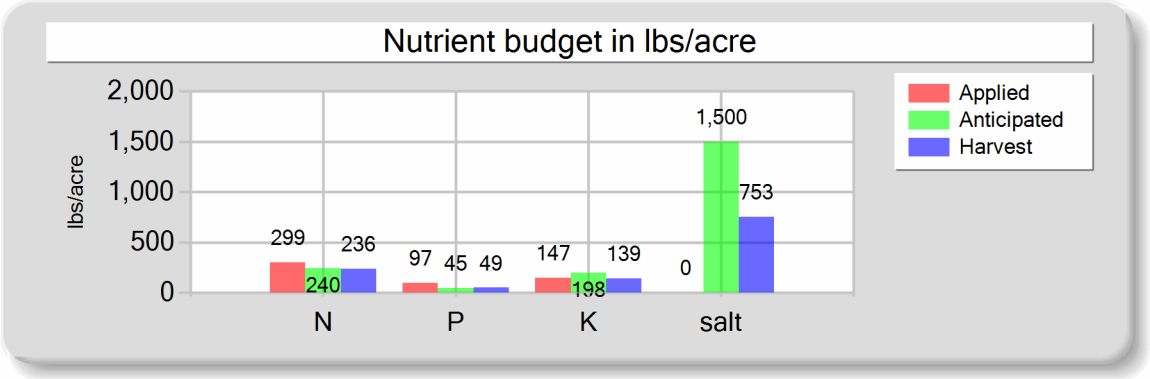
Fresh water applied	
5,567,378.00	*gallons*
205.03	*acre-inches*
3.06	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

East Walker - 06/06/2023: Corn, silage

Field name: East Walker | Crop: Corn, silage | Plant date: 06/06/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	50.00	0.00	0.00	0.00
Dry manure	242.35	96.63	146.50	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	299.35	96.63	146.50	0.00
Anticipated crop nutrient removal	240.00	45.00	198.00	1,500.00
Actual crop nutrient removal	235.80	49.35	138.92	753.11
Nutrient balance	63.55	47.27	7.58	-753.11
Applied to removed ratio	1.27	1.96	1.05	0.00

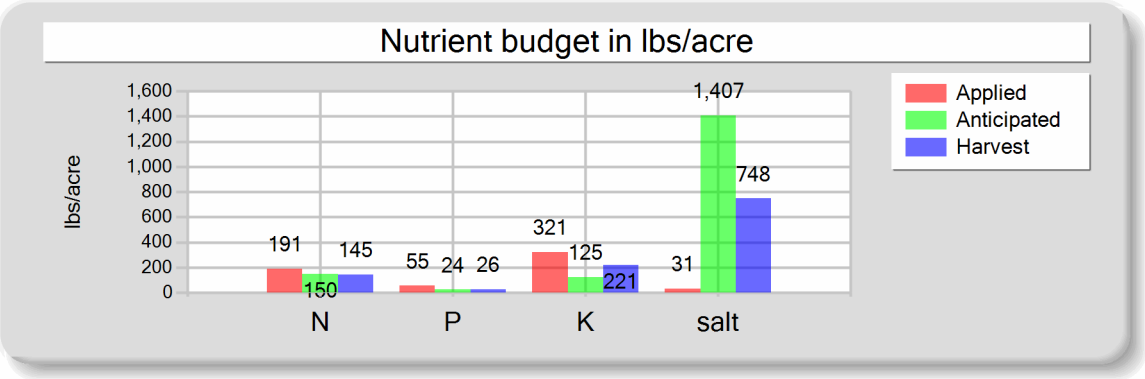
Fresh water applied	
65,270,198.00	*gallons*
2,403.68	*acre-inches*
35.88	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

HR North - 11/03/2022: Oats, silage-soft dough

Field name: HR North | Crop: Oats, silage-soft dough | Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	183.72	54.77	320.66	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	31.39
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	190.72	54.77	320.66	31.39
Anticipated crop nutrient removal	150.00	24.00	124.50	1,407.00
Actual crop nutrient removal	144.66	25.57	220.58	748.05
Nutrient balance	46.07	29.20	100.08	-716.66
Applied to removed ratio	1.32	2.14	1.45	0.04

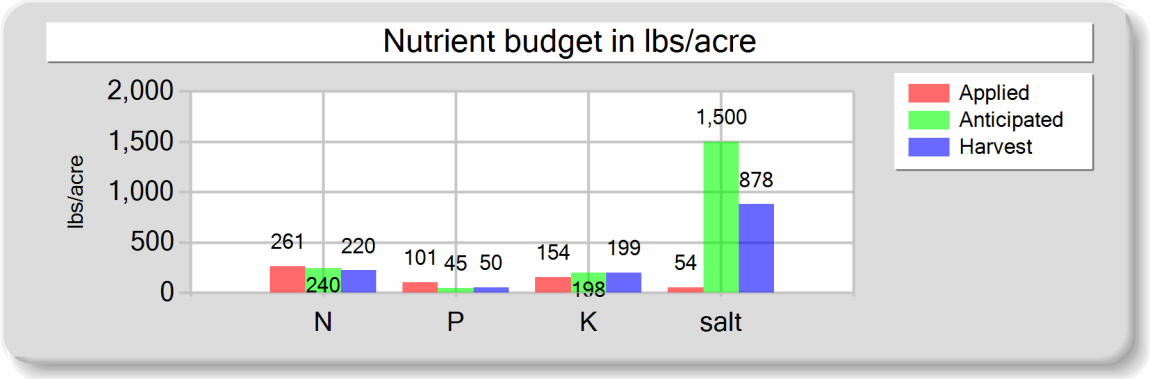
Fresh water applied	
1,504,840.00	*gallons*
55.42	*acre-inches*
3.46	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

HR North - 06/18/2023: Corn, silage

Field name: HR North | Crop: Corn, silage | Plant date: 06/18/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	254.16	101.34	153.64	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	53.99
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	261.16	101.34	153.64	53.99
Anticipated crop nutrient removal	240.00	45.00	198.00	1,500.00
Actual crop nutrient removal	220.32	49.75	199.00	877.71
Nutrient balance	40.85	51.59	-45.35	-823.73
Applied to removed ratio	1.19	2.04	0.77	0.06

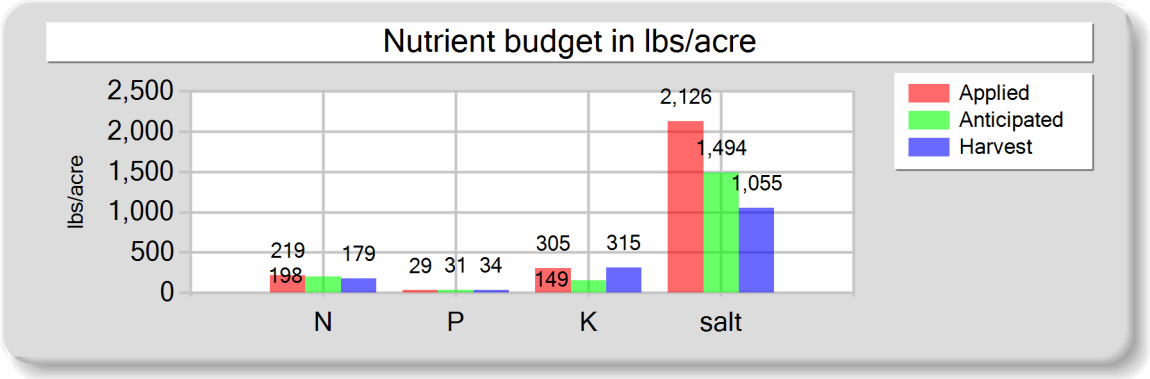
Fresh water applied	
15,105,369.00	*gallons*
556.28	*acre-inches*
34.77	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NRB - 11/12/2022: Wheat, silage, soft dough

Field name: NRB | Crop: Wheat, silage, soft dough | Plant date: 11/12/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	212.15	29.36	305.42	2,125.80
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	219.15	29.36	305.42	2,125.80
Anticipated crop nutrient removal	198.00	30.60	149.40	1,494.00
Actual crop nutrient removal	178.90	33.59	314.84	1,054.67
Nutrient balance	40.25	-4.23	-9.41	1,071.13
Applied to removed ratio	1.22	0.87	0.97	2.02

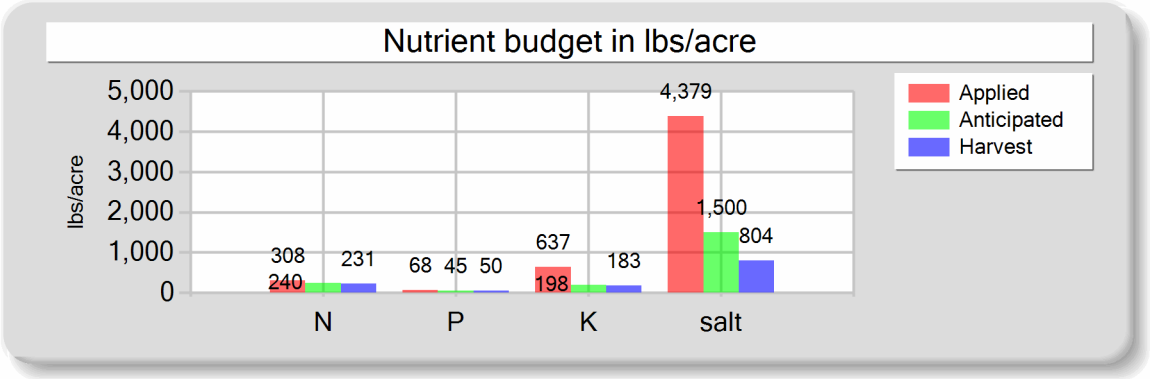
Fresh water applied
4,512,088.00 *gallons*
166.16 *acre-inches*
5.19 *inches/acre*

Process wastewater applied
2,625,510.00 *gallons*
96.69 *acre-inches*
3.02 *inches/acre*

Total harvests for the crop
1 *harvests*

NRB - 06/10/2023: Corn, silage

Field name: NRB | Crop: Corn, silage | Plant date: 06/10/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	301.08	68.22	637.25	4,272.63
Fresh water	0.00	0.00	0.00	106.20
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	308.08	68.22	637.25	4,378.83
Anticipated crop nutrient removal	240.00	45.00	198.00	1,500.00
Actual crop nutrient removal	231.50	50.33	182.85	803.52
Nutrient balance	76.59	17.90	454.40	3,575.31
Applied to removed ratio	1.33	1.36	3.49	5.45

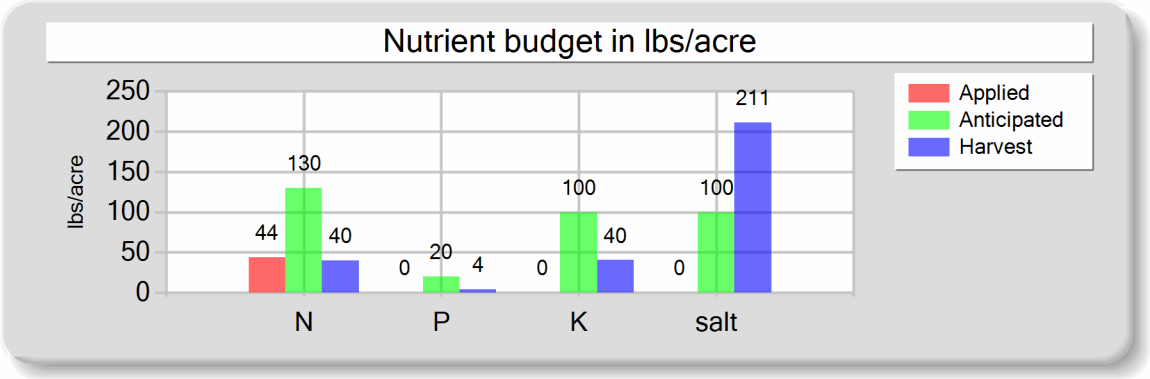
Fresh water applied	
32,433,990.00	*gallons*
1,194.43	*acre-inches*
37.33	*inches/acre*

Process wastewater applied	
2,519,800.00	*gallons*
92.80	*acre-inches*
2.90	*inches/acre*

Total harvests for the crop	
1	*harvests*

SM - 10/01/2018: Almond, in shell

Field name: SM | Crop: Almond, in shell | Plant date: 10/01/2018



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	30.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	44.00	0.00	0.00	0.00
Anticipated crop nutrient removal	130.00	20.00	100.00	100.00
Actual crop nutrient removal	40.15	4.27	40.34	211.41
Nutrient balance	3.85	-4.27	-40.34	-211.41
Applied to removed ratio	1.10	0.00	0.00	0.00

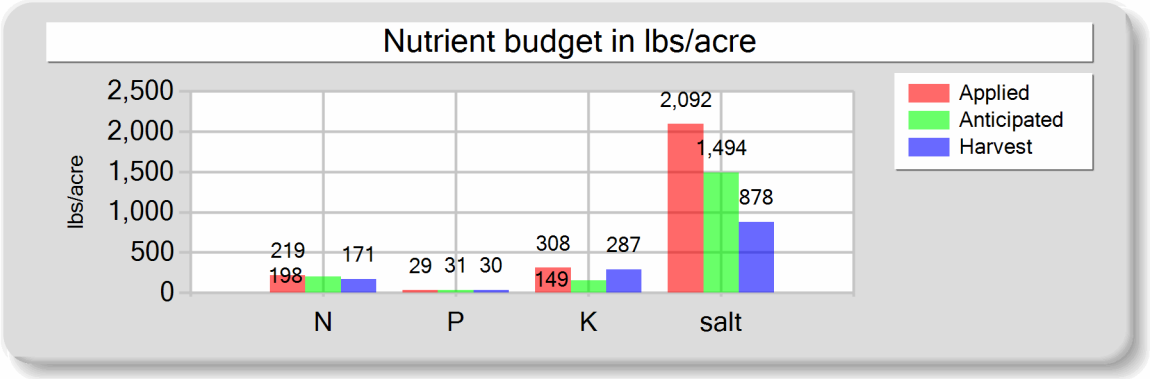
Fresh water applied	
20,705,968.00	*gallons*
762.53	*acre-inches*
31.77	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

SRB - 11/12/2022: Wheat, silage, soft dough

Field name: SRB | Crop: Wheat, silage, soft dough | Plant date: 11/12/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	212.00	28.90	307.50	2,092.43
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	219.00	28.90	307.50	2,092.43
Anticipated crop nutrient removal	198.00	30.60	149.40	1,494.00
Actual crop nutrient removal	170.57	30.48	286.70	878.25
Nutrient balance	48.43	-1.59	20.80	1,214.17
Applied to removed ratio	1.28	0.95	1.07	2.38

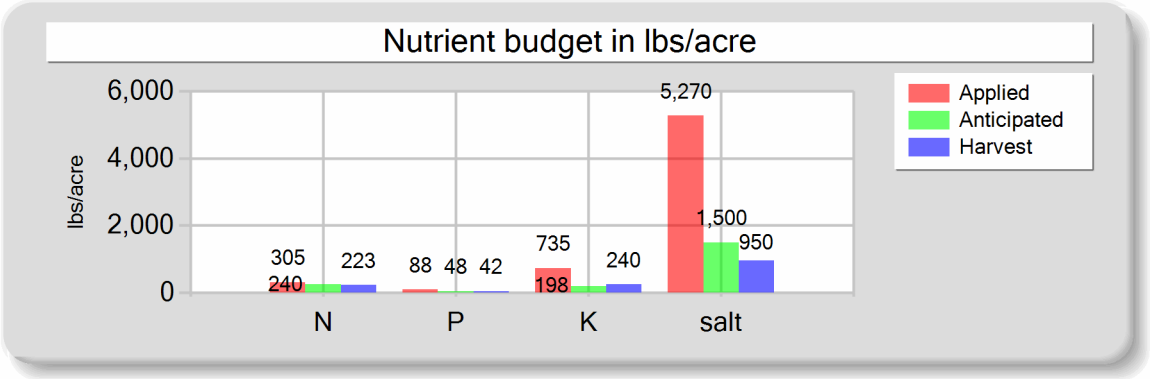
Fresh water applied	
7,176,240.00	*gallons*
264.28	*acre-inches*
8.01	*inches/acre*

Process wastewater applied	
2,534,973.00	*gallons*
93.35	*acre-inches*
2.83	*inches/acre*

Total harvests for the crop	
1	*harvests*

SRB - 06/19/2023: Corn, silage

Field name: SRB | Crop: Corn, silage | Plant date: 06/19/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	297.70	87.56	734.65	5,270.16
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	304.70	87.56	734.65	5,270.16
Anticipated crop nutrient removal	240.00	48.00	198.00	1,500.00
Actual crop nutrient removal	223.17	41.84	240.21	950.00
Nutrient balance	81.53	45.71	494.44	4,320.16
Applied to removed ratio	1.37	2.09	3.06	5.55

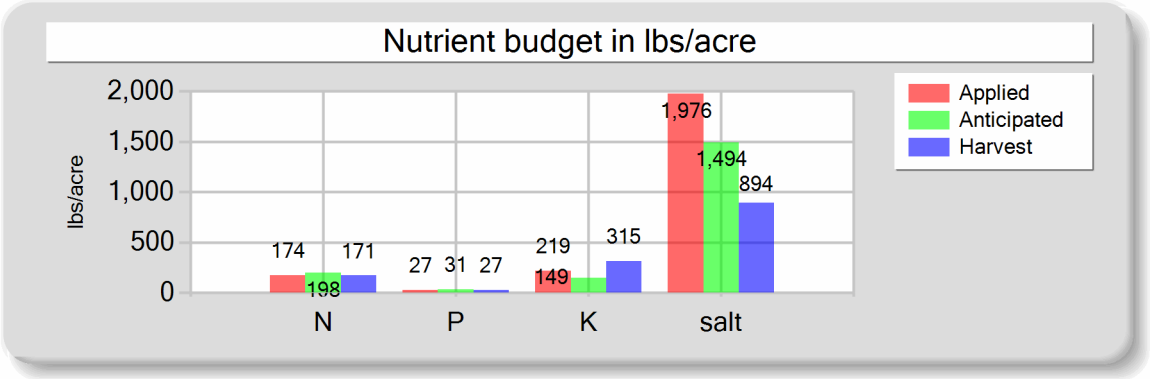
Fresh water applied	
33,340,928.00	*gallons*
1,227.83	*acre-inches*
37.21	*inches/acre*

Process wastewater applied	
3,297,573.00	*gallons*
121.44	*acre-inches*
3.68	*inches/acre*

Total harvests for the crop	
1	*harvests*

TD - 11/05/2022: Wheat, silage, soft dough

Field name: TD | Crop: Wheat, silage, soft dough | Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	167.33	27.32	219.32	1,975.79
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	174.33	27.32	219.32	1,975.79
Anticipated crop nutrient removal	198.00	30.60	149.40	1,494.00
Actual crop nutrient removal	170.58	27.27	315.16	894.13
Nutrient balance	3.75	0.05	-95.84	1,081.67
Applied to removed ratio	1.02	1.00	0.70	2.21

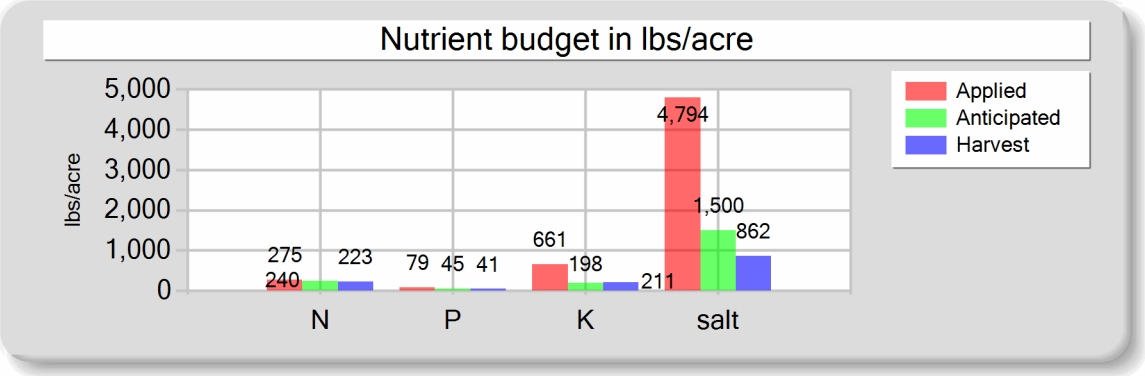
Fresh water applied	
2,665,060.00	*gallons*
98.15	*acre-inches*
8.18	*inches/acre*

Process wastewater applied	
1,359,410.00	*gallons*
50.06	*acre-inches*
4.17	*inches/acre*

Total harvests for the crop	
1	*harvests*

TD - 06/17/2023: Corn, silage

Field name: TD | Crop: Corn, silage | Plant date: 06/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	267.91	78.80	661.15	4,742.88
Fresh water	0.00	0.00	0.00	51.60
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	274.91	78.80	661.15	4,794.49
Anticipated crop nutrient removal	240.00	45.00	198.00	1,500.00
Actual crop nutrient removal	222.57	41.00	210.85	862.45
Nutrient balance	52.34	37.80	450.30	3,932.03
Applied to removed ratio	1.24	1.92	3.14	5.56

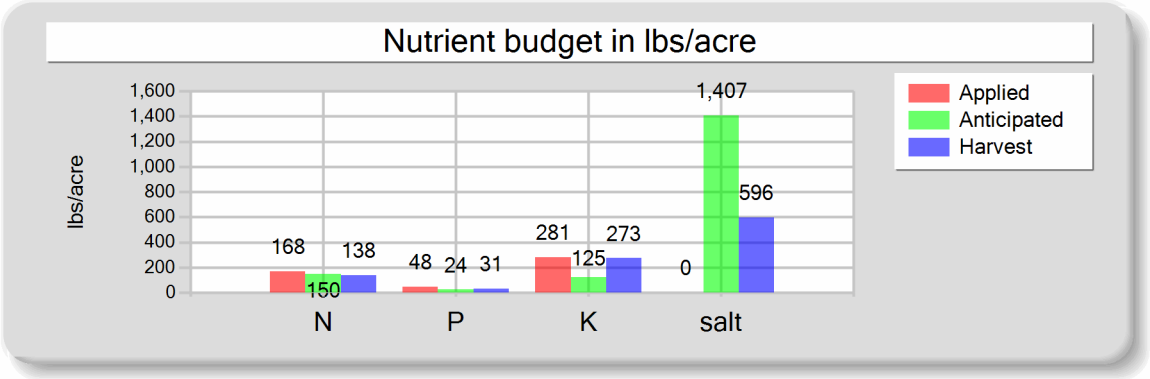
Fresh water applied	
11,045,598.00	*gallons*
406.77	*acre-inches*
33.90	*inches/acre*

Process wastewater applied	
1,079,146.00	*gallons*
39.74	*acre-inches*
3.31	*inches/acre*

Total harvests for the crop	
1	*harvests*

West Walker - 11/06/2022: Oats, silage-soft dough

Field name: West Walker | Crop: Oats, silage-soft dough | Plant date: 11/06/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	161.07	48.02	281.12	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	168.07	48.02	281.12	0.00
Anticipated crop nutrient removal	150.00	24.00	124.50	1,407.00
Actual crop nutrient removal	138.10	30.96	273.10	596.37
Nutrient balance	29.97	17.06	8.02	-596.37
Applied to removed ratio	1.22	1.55	1.03	0.00

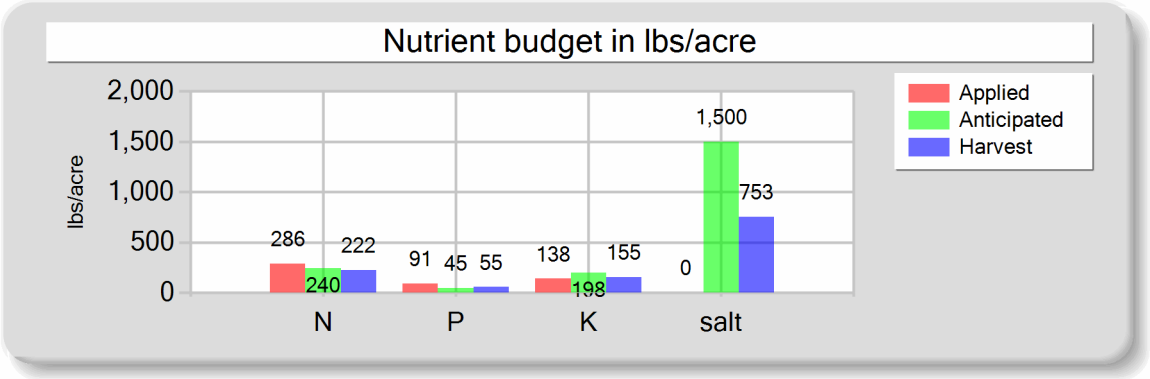
Fresh water applied	
4,414,800.00	*gallons*
162.58	*acre-inches*
2.23	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

West Walker - 06/17/2023: Corn, silage

Field name: West Walker | Crop: Corn, silage | Plant date: 06/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	50.00	0.00	0.00	0.00
Dry manure	228.80	91.22	138.31	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	285.80	91.22	138.31	0.00
Anticipated crop nutrient removal	240.00	45.00	198.00	1,500.00
Actual crop nutrient removal	222.13	54.65	155.14	752.76
Nutrient balance	63.67	36.57	-16.83	-752.76
Applied to removed ratio	1.29	1.67	0.89	0.00

Fresh water applied	
68,600,467.00	*gallons*
2,526.32	*acre-inches*
34.61	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Manure

Sample and source description: Manure

Sample date: 11/07/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 46.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,100.00	4,800.00	28,100.00							
DL	100.00	200.00	200.00							

Dry Manure

Sample and source description: Dry Manure

Sample date: 05/01/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 53.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	31,100.00	12,400.00	18,800.00							
DL	500.00	200.00	200.00							

Dry Manure

Sample and source description: Dry Manure

Sample date: 10/11/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 16.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,900.00	9,000.00	23,800.00							
DL	500.00	200.00	200.00							

B. PROCESS WASTEWATER ANALYSES

Lagoon

Sample and source description: Lagoon

Sample date: 11/07/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	177.00	68.10	0.00	0.00	28.90	232.00								3,150.00	2,090
DL	2.00	2.00	2.00	2.00	0.20	0.50								100.00	100

Lagoon

Sample and source description: Lagoon

Sample date: 03/08/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	846.00	463.00	0.00	0.00	99.30	1,310.00								10,800.00	7,200
DL	10.00	2.00	2.00	2.00	0.20	0.50								100.00	10

Lagoon

Sample and source description: Lagoon

Sample date: 05/01/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,430.00	597.00	0.00	0.00	387.00	1,720.00								6,760.00	4,490
DL	10.00	2.00	2.00	2.00	0.20	0.50								100.00	10

Lagoon

Sample and source description: Lagoon

Sample date: 08/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	357.00	352.00	0.00	0.00	105.00	881.00								9,520.00	6,320
DL	10.00	2.00	2.00	2.00	0.20	0.50								100.00	10

Lagoon

Sample and source description: Lagoon

Sample date: 11/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	805.00	615.00	0.00	0.00	156.00	1,580.00								12,500.00	8,320
DL	10.00	2.00	2.00	2.00	0.20	0.50								100.00	10

C. FRESH WATER ANALYSES

1

Domestic Well

Sample description: Domestic Well

Sample date: 12/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	39.10	0.00	39.10								1,050.00	
DL	0.20	0.40	0.20								1.00	

2

Domestic Well

Sample description: Domestic Well

Sample date: 12/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	49.00	0.00	49.00								1,120.00	
DL	0.20	0.20	0.20								1.00	

Turlock Irrigation District

Turlock Irrigation District

Irrigation Water

Sample description: Irrigation Water

Sample date: 09/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								49.00	0
DL	0.20	0.20	0.20								1.00	20

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

East Walker - 11/08/2022: Oats, silage-soft dough

Oat Silage

Sample and source description: Oat Silage

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,400.00	3,400.00	28,000.00		9.44
DL	500.00	200.00	200.00		0.05

East Walker - 06/06/2023: Corn, silage

East Walker - 06/06/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 08/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,900.00	2,700.00	7,600.00		4.12
DL	500.00	200.00	200.00		0.05

HR North - 11/03/2022: Oats, silage-soft dough

Oat Silage

Sample and source description: Oat Silage

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,100.00	3,200.00	27,600.00		9.36
DL	500.00	200.00	200.00		0.05

HR North - 06/18/2023: Corn, silage

Corn

Sample and source description: Corn

Sample date: 08/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,400.00	2,800.00	11,200.00		4.94
DL	500.00	200.00	200.00		0.05

NRB - 11/12/2022: Wheat, silage, soft dough

Wheat

Sample and source description: Wheat

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,900.00	4,300.00	40,300.00		13.50
DL	500.00	200.00	200.00		0.05

NRB - 06/10/2023: Corn, silage

Corn

Sample and source description: Corn

Sample date: 09/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,800.00	3,000.00	10,900.00		4.79
DL	500.00	200.00	200.00		0.05

SM - 10/01/2018: Almond, in shell

Almond

Sample and source description: Almond

Sample date: 10/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 6.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,700.00	2,200.00	20,800.00		10.90
DL	500.00	200.00	200.00		0.05

SRB - 11/12/2022: Wheat, silage, soft dough

Wheat

Sample and source description: Wheat

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,500.00	4,200.00	39,500.00		12.10
DL	500.00	200.00	200.00		0.05

SRB - 06/19/2023: Corn, silage

Corn

Sample and source description: Corn

Sample date: 09/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,400.00	2,700.00	15,500.00		6.13
DL	500.00	200.00	200.00		0.05

TD - 11/05/2022: Wheat, silage, soft dough

Wheat

Sample and source description: Wheat

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,900.00	4,300.00	49,700.00		14.10
DL	500.00	200.00	200.00		0.05

TD - 06/17/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,200.00	2,800.00	14,400.00		5.89
DL	500.00	200.00	200.00		0.05

West Walker - 11/06/2022: Oats, silage-soft dough

Oat Silage

Sample and source description: Oat Silage

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 79.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,300.00	5,000.00	44,100.00		9.63
DL	500.00	200.00	200.00		0.05

West Walker - 06/17/2023: Corn, silage

Corn

Sample and source description: Corn

Sample date: 08/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,600.00	3,100.00	8,800.00		4.27
DL	500.00	200.00	200.00		0.05

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

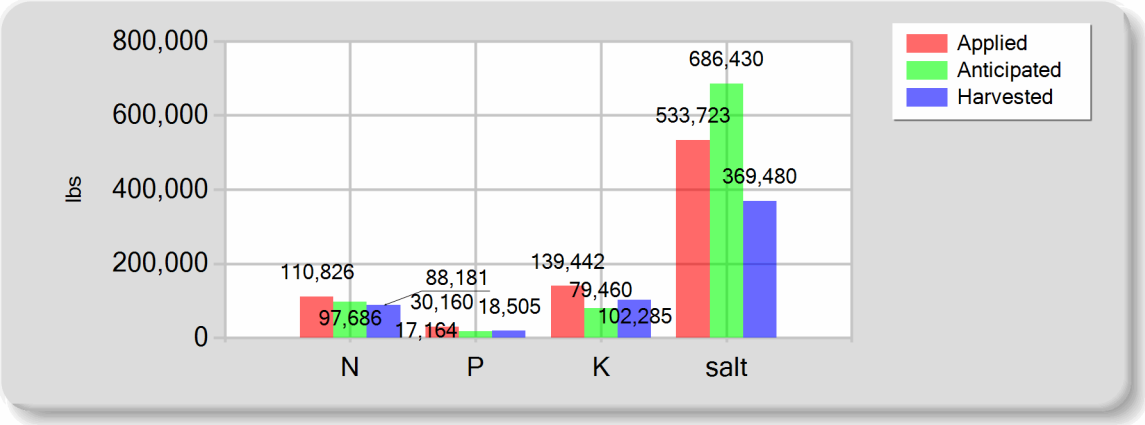
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

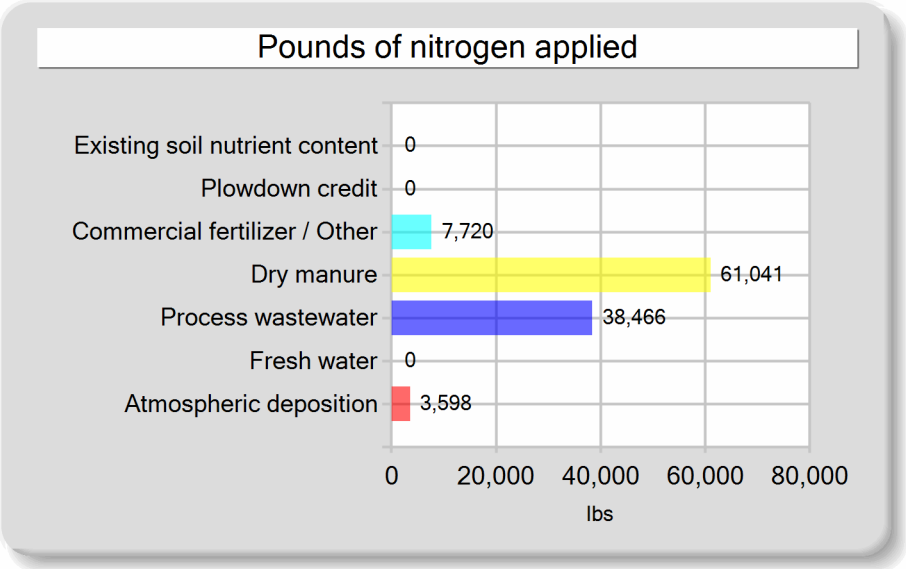
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

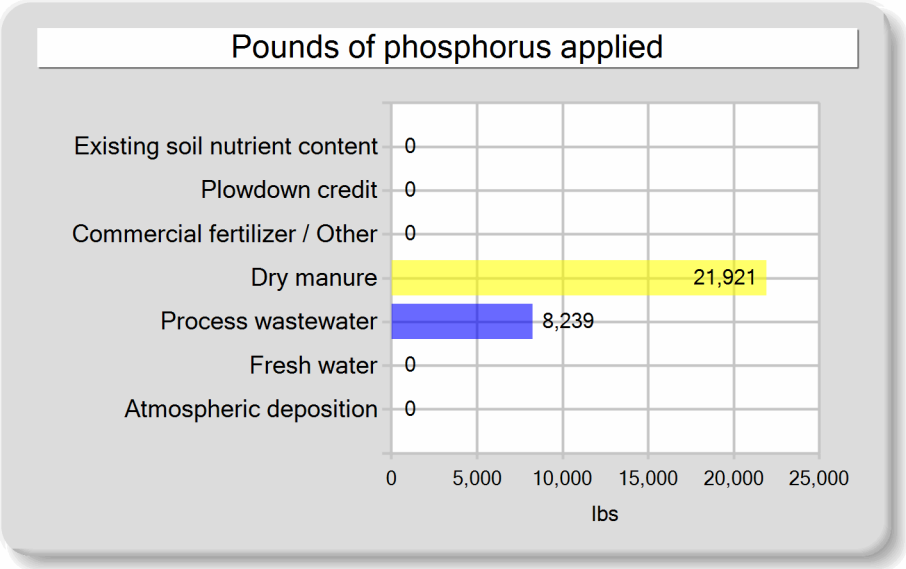
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	7,720.00	0.00	0.00	0.00
Dry manure	61,041.43	21,920.69	64,319.79	0.00
Process wastewater	38,466.45	8,239.01	75,122.38	528,339.45
Fresh water	0.00	0.00	0.00	5,383.76
Atmospheric deposition	3,598.00	0.00	0.00	0.00
Total nutrients applied	110,825.88	30,159.69	139,442.17	533,723.21
Anticipated crop nutrient removal	97,686.00	17,164.20	79,459.80	686,430.00
Actual crop nutrient removal	88,181.26	18,505.12	102,285.14	369,480.10
Nutrient balance	22,644.62	11,654.58	37,157.03	164,243.11
Applied to removed ratio	1.26	1.63	1.36	1.44

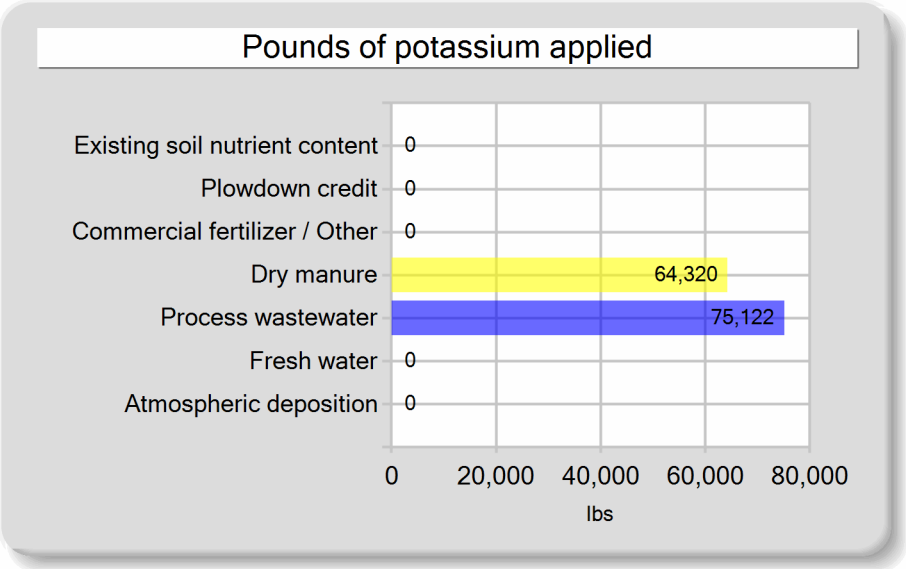
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

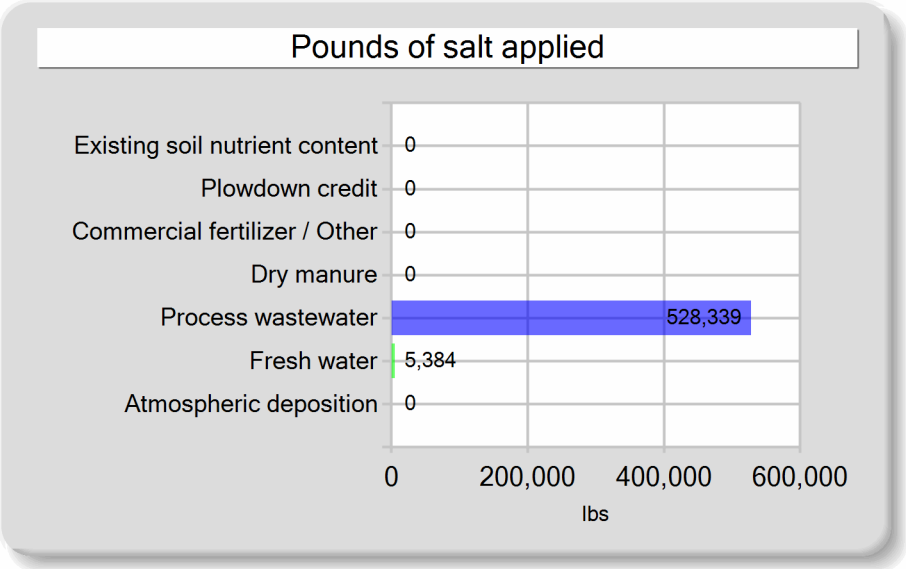


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Wells~ IW1, IW2 were out of service in 2023.
2nd Qtr. wastewater results not utilized - erroneous data set.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Steven Meirinho	Sam
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
	Steven Meirinho
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6/4/24	Sam
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

ENVIRONMENTAL AGRICULTURAL
Analytical Chemists

September 29, 2023

Sentry Ag Services
Attn: Monique Baldivez
P.O. Box 7750
Visalia, CA 93290

Lab No. : **VI 2346144**
Customer No. : **4019696**
Reference : **3148**

Laboratory Report

Introduction: This report package contains a total of 3 pages divided into 3 sections:

Case Narrative	(1 page)	: An overview of the work performed at FGL.
Sample Results	(1 page)	: Results for each sample submitted.
Quality Control	(1 page)	: Supporting Quality Control (QC) results.

Case Narrative

This Case Narrative pertains to the following samples:

Sample Description	Date Sampled	Date Received	FGL Lab No.	Matrix
Canal	09/11/2023	09/11/2023	VI 2346144-001	AGW

Sampling and Receipt Information:

The Sample was received in acceptable condition and within temperature requirements, unless noted on the Condition Upon Receipt (CUR) form. The Sample was received, prepared and analyzed within the method specified holding times. All samples arrived on ice. All samples were checked for pH if acid or base preservation is required (except for VOAs). For details of sample receipt information, please see the associated Chain of Custody and Condition Upon Receipt Form.

Quality Control: All samples were prepared and analyzed according to established quality control criteria. Any exceptions are noted in the Quality Control Section of this report.

Test Summary	
EPA 351.2	Preparation and analysis performed by FGL-Santa Paula (FGL-SP ELAP# 1573)
SM 2540 C	Preparation and analysis performed by FGL-Santa Paula (FGL-SP ELAP# 1573)
SM 4500-H+B	Preparation and analysis performed by FGL-Santa Paula (FGL-SP ELAP# 1573)
SM 4500-NO3 F	Preparation and analysis performed by FGL-Santa Paula (FGL-SP ELAP# 1573)

Certification: I certify that this data package is in compliance with ELAP standards, both technically and for completeness, except for any conditions listed above and in the QC Section. Release of the data contained in this data package is authorized by the Laboratory Director or his designee, as verified by the following electronic signature. This report shall not be reproduced except in full, without the written approval of the laboratory.

KD: EHB

Approved By **Kelly A. Dunnahoo, B.S.** Digitally signed by Kelly A. Dunnahoo, B.S.
Title: Laboratory Director
Date: 2023-09-29

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9415 W. Goshen Avenue
Visalia, CA 93291
TEL: (559)734-9473
FAX: (559)734-8435
CA ELAP Certification No. 2810

September 29, 2023

Sentry Ag Services
Attn: Monique Baldivez
P.O. Box 7750
Visalia, CA 93290

Description : Canal
Project : Turlock Irrigation District

Lab No. : VI 2346144-001
Customer No. : 4019696
Reference : 3148
Sampled On : September 11, 2023 at 10:05
Sampled By : Darin
Received On : September 11, 2023 at 15:20
Matrix : Ag Water

Sample Results - Inorganic

Constituent	Result	RL	Units	Note	Dil.	DQF	Sample Preparation			Sample Analysis			
Dairy Analysis							Date	Time	Who	Method	Date	Time	Who
Nitrogen, Total Kjeldahl	ND	0.5	mg/L		1	Ul	09/21/2023	09:23	sta	EPA 351.2	09/25/2023	20:50	lcr
Nitrate Nitrogen	ND	0.4	mg/L		1	U	09/12/2023	12:35	lfs	SM 4500-NO3 F	09/12/2023	14:55	lfs
Nitrogen, Total as Nitrogen	ND	0.5	mg/L		1	Ul	09/21/2023	09:23	sta	Calc.	09/25/2023	20:50	lcr
Nitrate + Nitrite as N	ND	0.4	mg/L		1	U	09/12/2023	12:35	lfs	SM 4500-NO3 F	09/12/2023	14:55	lfs
Kjeldahl Nitrogen	ND	0.5	mg/L		1	Ul	09/21/2023	09:23	sta	EPA 351.2	09/25/2023	20:50	lcr
Conductivity	49	1	umhos/cm		1		09/15/2023	09:09	krh	SM 4500-H+B	09/15/2023	10:30	krh
Solids, Total Dissolved (TDS)	ND	20	mg/L		1	J	09/13/2023	10:20	ctl	SM 2540 C	09/14/2023	11:40	ctl

DQF Flags Definition:
U Constituent results were non-detect.
l The MS/MSD did not meet QC criteria.
J Reported value is estimated; detected at a concentration below the RL and above the laboratory MDL.

ND=Non-Detected, RL=Reporting Level , Dil.=Dilution

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FAX: (559)734-8435
CA ELAP Certification No. 2810

September 29, 2023
Sentry Ag Service

Lab No. : VI 2346144
Customer No. : 4019696

Quality Control - Wet Chem

Constituent	Method	Date/ID	Type	Units	Conc.	QC Data	DQO	Note
Wet Chem								
E. C.	2320B	(VI 2346608-002)	Dup	umhos/cm		0.2%	5	
Solids, Total Dissolved	2540CE	09/13/2023:210246CTL	Blank	mg/L		ND	<20	
			LCS	mg/L	991.5	101%	90-110	
		(VI 2346141-005)	Dup	mg/L		1.75%	5	
		(VI 2346141-005)	Dup	mg/L		2.07%	5	
Nitrogen, Total Kjeldahl	351.2	09/21/2023:210595STA	Blank	mg/L		ND	<0.5	
			LCS	mg/L	12.00	91.7%	73-124	
			MS	mg/L	12.00	89.3%	90-110	435
		(STK2352400-001)	MSD	mg/L	12.00	88.1%	90-110	435
			MSRPD	mg/L		1.2%	≤20	
			MS	mg/L	12.00	89.7%	90-110	435
		(STK2352400-004)	MSD	mg/L	12.00	93.6%	90-110	
			MSRPD	mg/L		3.8%	≤20	
Nitrate + Nitrite as N	4500NO3F	09/12/2023:210228LFS	Blank	mg/L		ND	<0.4	
			LCS	mg/L	11.22	98.4%	80-120	
			MS	mg/L	5.609	101%	66-125	
		(CH 2377709-001)	MSD	mg/L	5.609	102%	66-125	
			MSRPD	mg/L		1.1%	≤30.4	
Nitrate Nitrogen	4500NO3F	09/12/2023:210228LFS	Blank	mg/L		ND	<0.4	
			LCS	mg/L	11.22	98.4%	80-120	
			MS	mg/L	5.609	101%	66-125	
		(CH 2377709-001)	MSD	mg/L	5.609	102%	66-125	
			MSRPD	mg/L		1.1%	≤30.4	

Definition

Blank : Method Blank - Prepared to verify that the preparation process is not contributing contamination to the samples.
DQO : Data Quality Objective - This is the criteria against which the quality control data is compared.
Dup : Duplicate Sample - A random sample with each batch is prepared and analyzed in duplicate. The relative percent difference is an indication of precision for the preparation and analysis.
LCS : Laboratory Control Standard/Sample - Prepared to verify that the preparation process is not affecting analyte recovery.
MS : Matrix Spikes - A random sample is spiked with a known amount of analyte. The recoveries are an indication of how that sample matrix affects analyte recovery.
MSD : Matrix Spike Duplicate of MS/MSD pair - A random sample duplicate is spiked with a known amount of analyted. The recoveries are an indication of how that sample matrix affects analyte recovery.
MSRPD : MS/MSD Relative Percent Difference (RPD) - The MS relative percent difference is an indication of precision for the preparation and analysis.
ND : Non-detect - Result was below the DQO listed for the analyte.

Explanation

435 : Sample matrix may be affecting this analyte. Data was accepted based on the LCS or CCV recovery.

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CA ELAP Certification No. 2810

Laboratory Analysis Work Order 23461144

3148

SITE NAME: Turlock Irrigation District

LABORATORY: VT | FGL 4-19696

Billing: Sentry Ag Services, LLC
P.O. Box 7750, Visalia, CA 93290

Authorized Copy Release to:
labs@sentryagservices.com

ANALYSIS TO BE COMPLETED

Irrigation/Ground Water (ELAP Standards)

- W1 EC, NO_3N (Dom)
- W2 EC, NO_3N, TDS, TN (Irr)
- W3 NH_4-N (Ammonium)
- W4 EC, NO_3N, Ca, Mg, Na, HCO_3, CO_3, SO_4S, Cl, TDS (Dom, GM)
- W5 EC, NO_3N, TDS, TN, Ca, Mg, Na, HCO_3, CO_3, SO_4S, Cl (Irr, GM)
- W6 NO_3N, NO_2 (Dom ILRP, Annually)
- W7 Ca, Mg, Na, K, HCO_3, CO_3, SO_4, Cl + Lab Filtering (GWM)
- W8 Other: _____

RCL 4.1 C Tot 17/407

Plant Tissue

- P1 TN, NO_3N, PO_4P, K (Mid Season - Wheat)
- P2 TN, P, K (Mid-season - Corn)
- P3 TN, TP, TK, Ash, %M (At Harvest)
- P4 TN, %M
- P5 % Moisture
- P6 NIR
- P7 Other: _____

Process Waste Water (lagoon)

- L1 EC, NH_4N, TKN, TP, TK, TDS (Quarterly)
- L2 EC, NO_3N, NH_4N, TKN, TP, TK, TDS, pH (Annually)
- L3 Ca, Mg, Na, HCO_3,CO_3, SO_4S, Cl (Biennially)
- L4 Other: _____

Manure

- M1 TN, TP, TK, %M (2/year)
- M2 TN, TP, K, %M, Ca, Mg, Na, S, Cl, ash (Biennially)
- M3 Other: _____

Soil

- S1 SP%, pH, EC, Ca, Mg, Na, K, ESP, LP, B, NO_3N, PO_4P, K-AA, Zn, Mn, Fe, Cu, SO_4S
- S2 S1 + CEC, CaCO3, OM, C:N, TN
- S3 NO_3N, NH_4N
- S4 Other: _____

	Sample ID	Description	Analysis	Date/Time	Sampled by	SAS USE ONLY: FIELD TESTS NH_3N*	pH	Temp
1	Canal	irrigation water	W2	9/11/23 10:05	Darin			
2								
3								
4								
5								
6								
7								
8								
9								
10								
11								
12								

* Field Test of ammonium nitrogen may only be made by a trained technician. Positive test to be analyzed for ammonium nitrogen by the laboratory.

All samples are to follow the procedures noted in the Sampling & Analysis Plan of the NMP and the RWQCB specifications. Any samples taken outside of these procedures shall provide the procedures on the notes below. Additionally, if any preservatives are used in the collections or processing of sampes, please note below.

NOTES:

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time
1st	[illegible]	SAS		9/11/23 3:20 pm
2nd	[illegible]	FGL	9/11/23 1520	
3rd	[illegible]			9/11/23 1730
4th	[illegible]		9/11/23 1730	

LABORATORY USE ONLY
Logged In By: _____ Total Samples: _____ Laboratory No.: _____

GLS 9/12/23
mc 12:17

Inter-Laboratory Condition Upon Receipt (Attach to COC)

Sample Receipt at: STK CC CH VI

1. Number of ice chests/packages received: 1 Shipping tracking # GTC
2. Were samples received in a chilled condition? Temps: 2 / 4 / 1 / 0 / ____ / ____ / ____

Surface water SWTR bact samples: A sample that has a temperature upon receipt of >10° C, whether iced or not, should be flagged unless the time since sample collection has been less than two hours.

3.	Do the number of bottles received agree with the COC?	Yes	No	N/A
4.	Were samples received intact? (i.e. no broken bottles, leaks etc.)	Yes	No	
5.	VOAs checked for Headspace?	Yes	No	N/A
6.	Were sample custody seals intact?	Yes	No	N/A
7.	If required, was sample split for pH analysis?	Yes	No	N/A
8.	Were all analyses within holding times at time of receipt?	Yes	No	
9.	Verify sample date, time and sampler name	Yes	No	

Sign and date the COC, place in a ziplock and ~~put in~~ the same ice chest as the samples.
Sample Receipt Review completed by (initials): AH

Sample Receipt at SP:

1. Were samples received in a chilled condition? Temps: 2 / ____ / ____ / ____ / ____
   Acceptable is above freezing to 6 C. If many packages are received at one time check for tests/H.T.'s/rushes/
2. Shipping tracking numbers: 560100233 243 255

3.	Do the number of bottles received agree with the COC?	Yes	No	N/A
4.	Were samples received intact? (i.e. no broken bottles, leaks etc.)	Yes	No	
5.	Were sample custody seals intact?	Yes	No	N/A

Sign and date the COC, obtain LIMS sample numbers, select methods/tests and print labels.

Sample Verification, Labeling and Distribution:

1.	Were all requested analyses understood and acceptable?	Yes	No		
2.	Did bottle labels correspond with the client's ID's?	Yes	No		
3.	Were all bottles requiring sample preservation properly preserved? [Exception: Oil & Grease, VOA and CrVI verified in lab]	Yes	No	N/A	FGL
4.	VOAs checked for Headspace?	Yes	No	N/A	
5.	Have rush or project due dates been checked and accepted?	Yes	No	N/A	
6.	Were all analyses within holding times at time of receipt?	Yes	No		

Attach labels to the containers and include a copy of the COC for lab delivery.
Sample Receipt, Login and Verification completed by (initials): MC

Discrepancy Documentation:

Any items above which are "No" or do not meet specifications (i.e. temps) must be resolved.

1. Person Contacted:_____ Phone Number:_____
   Initiated By:_____ Date:_____
   Problem:
   Resolution:

2. Person Contacted:_____
   Initiated By:_____
   Problem:
   Resolution:

(Please use the back of this sheet for additional contacts)

(4019696)
Sentry Ag Service
VI 2346144
cda 09/12/2023 07:49:32
VI 2346144

ENVIRONMENTAL AGRICULTURAL
Analytical Chemists

December 18, 2023

Sentry Ag Services
Attn: Monique Baldivez
P.O. Box 7750
Visalia, CA 93290

Lab No. : **VI 2348406**
Customer No. : **4019696**
Reference : **3486**

Laboratory Report

Introduction: This report package contains a total of 4 pages divided into 3 sections:

Case Narrative	(1 page)	: An overview of the work performed at FGL.
Sample Results	(2 pages)	: Results for each sample submitted.
Quality Control	(1 page)	: Supporting Quality Control (QC) results.

Case Narrative

This Case Narrative pertains to the following samples:

Sample Description	Date Sampled	Date Received	FGL Lab No.	Matrix
Dom 1	12/11/2023	12/11/2023	VI 2348406-001	DW
Dom 2	12/11/2023	12/11/2023	VI 2348406-002	DW

Sampling and Receipt Information:

All samples were received in acceptable condition and within temperature requirements, unless noted on the Condition Upon Receipt (CUR) form. All samples were received, prepared and analyzed within the method specified holding times. All samples arrived on ice. All samples were checked for pH if acid or base preservation is required (except for VOAs). For details of sample receipt information, please see the associated Chain of Custody and Condition Upon Receipt Form.

Quality Control: All samples were prepared and analyzed according to established quality control criteria. Any exceptions are noted in the Quality Control Section of this report.

Test Summary	
SM 4500-H+B	Preparation and analysis performed by FGL-Santa Paula (FGL-SP ELAP# 1573)
SM 4500-NO3 F	Preparation and analysis performed by FGL-Santa Paula (FGL-SP ELAP# 1573)

Certification: I certify that this data package is in compliance with ELAP standards, both technically and for completeness, except for any conditions listed above and in the QC Section. Release of the data contained in this data package is authorized by the Laboratory Director or his designee, as verified by the following electronic signature. This report shall not be reproduced except in full, without the written approval of the laboratory.

KD: JRD

Approved By **Kelly A. Dunnahoo, B.S.** Digitally signed by Kelly A. Dunnahoo, B.S.
Title: Laboratory Director
Date: 2023-12-18

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December 18, 2023

Sentry Ag Services
Attn: Monique Baldivez
P.O. Box 7750
Visalia, CA 93290

Description : Dom 1
Project : Joe Meirinho & Son Dairy

Lab No. : VI 2348406-001
Customer No. : 4019696
Reference : 3486
Sampled On : December 11, 2023 at 12:20
Sampled By : Darin
Received On : December 11, 2023 at 15:20
Matrix : Drinking Water

Sample Results - Inorganic

Constituent	Result	RL	Units	MCL/AL	Dil.	DQF	Sample Preparation			Sample Analysis			
Dairy Analysis							Date	Time	Who	Method	Date	Time	Who
Nitrate Nitrogen	39.1	0.4	mg/L	10	1		12/13/2023	09:00	lfs	SM 4500-NO3 F	12/13/2023	09:33	lfs
Conductivity	1050	1	umhos/cm	1600[2]	1		12/15/2023	08:07	krh	SM 4500-H+B	12/15/2023	09:24	krh

DQF Flags Definition:

ND=Non-Detected, RL=Reporting Level , Dil.=Dilution
MCL = Maximum Contamination Level. 2 - Secondary Standard. 3 - CDPH Notification Level. AL = Regulatory Action Level.

Page 2 of 4

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December 18, 2023

Sentry Ag Services
Attn: Monique Baldivez
P.O. Box 7750
Visalia, CA 93290

Description : Dom 2
Project : Joe Meirinho & Son Dairy

Lab No. : VI 2348406-002
Customer No. : 4019696
Reference : 3486
Sampled On : December 11, 2023 at 12:22
Sampled By : Darin
Received On : December 11, 2023 at 15:20
Matrix : Drinking Water

Sample Results - Inorganic

Constituent	Result	RL	Units	MCL/AL	Dil.	DQF	Sample Preparation			Sample Analysis			
Dairy Analysis							Date	Time	Who	Method	Date	Time	Who
Nitrate Nitrogen	49.0	0.8*	mg/L	10	2		12/13/2023	09:00	lfs	SM 4500-NO3 F	12/13/2023	10:29	lfs
Conductivity	1120	1	umhos/cm	1600[2]	1		12/15/2023	08:07	krh	SM 4500-H+B	12/15/2023	09:27	krh

DQF Flags Definition:

ND=Non-Detected, RL=Reporting Level * RL adjusted for dilution, Dil.=Dilution
MCL = Maximum Contamination Level. 2 - Secondary Standard. 3 - CDPH Notification Level. AL = Regulatory Action Level.

Page 3 of 4

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December 18, 2023
Sentry Ag Service

Lab No. : VI 2348406
Customer No. : 4019696

Quality Control - Wet Chem

Constituent	Method	Date/ID	Type	Units	Conc.	QC Data	DQO	Note
Wet Chem								
E. C.	2320B	(SP 2320577-001)	Dup	umhos/cm		0.1%	5	
Nitrate Nitrogen	4500NO3F	12/13/2023:214023LFS	Blank	mg/L		ND	<0.4	
			LCS	mg/L	11.22	98.1%	80-120	
			MS	mg/L	5.609	96.7%	66-125	
		(VI 2348379-001)	MSD	mg/L	5.609	96.1%	66-125	
			MSRPD	mg/L		0.3%	≤30.4	

Definition

Blank : Method Blank - Prepared to verify that the preparation process is not contributing contamination to the samples.

DQO : Data Quality Objective - This is the criteria against which the quality control data is compared.

Dup : Duplicate Sample - A random sample with each batch is prepared and analyzed in duplicate. The relative percent difference is an indication of precision for the preparation and analysis.

LCS : Laboratory Control Standard/Sample - Prepared to verify that the preparation process is not affecting analyte recovery.

MS : Matrix Spikes - A random sample is spiked with a known amount of analyte. The recoveries are an indication of how that sample matrix affects analyte recovery.

MSD : Matrix Spike Duplicate of MS/MSD pair - A random sample duplicate is spiked with a known amount of analyted. The recoveries are an indication of how that sample matrix affects analyte recovery.

MSRPD : MS/MSD Relative Percent Difference (RPD) - The MS relative percent difference is an indication of precision for the preparation and analysis.

ND : Non-detect - Result was below the DQO listed for the analyte.

Page 4 of 4

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Laboratory Analysis Work Order

3486

2348406

SITE NAME: Joe Meirinho + Son Dairy

LABORATORY: ~~VT~~ FGL 4-19696

Billing: Sentry Ag Services, LLC
P.O. Box 7750, Visalia, CA 93290

Authorized Copy Release to:
labs@sentryagservices.com

ANALYSIS TO BE COMPLETED

Irrigation/Ground Water (ELAP Standards)

- W1 EC, NO_3N (Dom)
- W2 EC, NO_3N, TDS, TN (Irr)
- W3 NH_4-N (Ammonium)
- W4 EC, NO_3N, Ca, Mg, Na, HCO_3, CO_3, SO_4S, Cl, TDS (Dom, GM)
- W5 EC, NO_3N, TDS, TN, Ca, Mg, Na, HCO_3, CO_3, SO_4S, Cl (Irr, GM)
- W6 NO_3N, NO_2 (Dom ILRP, Annually)
- W7 Ca, Mg, Na, K, HCO_3, CO_3, SO_4, Cl + Lab Filtering (GWM)
- W8 Other: _____

Plant Tissue

- P1 TN, NO_3N, PO_4P, K (Mid Season - Wheat)
- P2 TN, P, K (Mid-season - Corn)
- P3 TN, TP, TK, Ash, %M (At Harvest)
- P4 TN, %M
- P5 % Moisture
- P6 NIR
- P7 Other: _____

Process Waste Water (lagoon)

- L1 EC, NH_4N, TKN, TP, TK, TDS (Quarterly)
- L2 EC, NO_3N, NH_4N, TKN, TP, TK, TDS, pH (Annually)
- L3 Ca, Mg, Na, HCO_3,CO_3, SO_4S, Cl (Biennially)
- L4 Other: _____

Manure

- M1 TN, TP, TK, %M (2/year)
- M2 TN, TP, K, %M, Ca, Mg, Na, S, Cl, ash (Biennially)
- M3 Other: _____

Soil

- S1 SP%, pH, EC, Ca, Mg, Na, K, ESP, LP, B, NO_3N, PO_4P, K-AA, Zn, Mn, Fe, Cu, SO_4S
- S2 S1 + CEC, CaCO3, OM, C:N, TN
- S3 NO_3N, NH_4N
- S4 Other: _____

	Sample ID	Description	Analysis	Date/Time	Sampled by	SAS USE ONLY: FIELD TESTS NH_3N *	pH	Temp
1	Dom 1	domestic well	W1	12/11/23 12:22pm	Darin			
2	Dom 2	domestic well	W1	12/11/23 12:22pm	Darin			
3								
4								
5								
6								
7								
8								
9								
10								
11								
12								

* Field Test of ammonium nitrogen may only be made by a trained technician. Positive test to be analyzed for ammonium nitrogen by the laboratory.

All samples are to follow the procedures noted in the Sampling & Analysis Plan of the NMP and the RWQCB specifications. Any samples taken outside of these procedures shall provide the procedures on the notes below. Additionally, if any preservatives are used in the collections or processing of sampes, please note below.

NOTES: R01 9.7°C ID# 77407

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time
1st	[signature]	SAS		12/11/23 3:20pm
2nd	SRO	FGL	12·11·23 1520	
3rd	SRO	FGL		12·11·23 1736
4th		GLS	12·11·23 1730	

LABORATORY USE ONLY
Logged In By: _____ **Total Samples:** _____ **Laboratory No.:** _____

GLS 12/12/23
MC 1013

Inter-Laboratory Condition Upon Receipt (Attach to COC)

Sample Receipt at: CC CH STK VI

1. Number of ice chests/packages received: 1 Shipping tracking #(s): OTC
2. Temp IR Gun ID #: TH407
3. Were samples received on ice? Yes No Temps: 9.7°C ____ / ____ / ____ / ____
   Surface water SWTR bact samples: A sample that has a temperature upon receipt of >10° C, whether iced or not, should be flagged unless the time since sample collection has been less than two hours.
4. Do the number of bottles received agree with the COC? ~~Yes~~ No N/A
5. Were samples received intact? (i.e. no broken bottles, leaks etc.) Yes No
6. VOAs checked for Headspace? Yes No N/A
7. Were all analyses within holding times at time of receipt? Yes No
8. Verify sample date, time and sampler name Yes No

Sign and date the COC, place in a ziplock and put in the same ice chest as the samples.
Sample Receipt Review completed by (initials): SRO

Sample Receipt at SP:

1. Number of ice chests/packages received: 5 Shipping tracking #(s): 5606126649 918 289 635 912
2. Temp IR Gun ID #: TH261
3. Were samples received on ice? ~~Yes~~ No Temps: 1c / ____ / ____ / ____ / ____
   Acceptable is above freezing to 6°C. If many packages are received at one time check for tests/H.T.'s/rushes/
4. Do the number of bottles received agree with the COC? ~~Yes~~ No N/A
5. Were samples received intact? (i.e. no broken bottles, leaks etc.) ~~Yes~~ No

Sign and date the COC, obtain LIMS sample numbers, select methods/tests and print labels.

Sample Verification, Labeling and Distribution:

1. Were all requested analyses understood and acceptable? Yes No
2. Did bottle labels correspond with the client's ID's? Yes No
3. Were all bottles requiring sample preservation properly preserved? Yes No N/A FGL
   [Exception: Oil & Grease, VOA and CrVI verified in lab]
4. VOAs checked for Headspace? Yes No N/A
5. Have rush or project due dates been checked and accepted? Yes No ~~N/A~~
6. Were all analyses within holding times at time of receipt? Yes No

Attach labels to the containers and include a copy of the COC for lab delivery.
Sample Receipt, Login and Verification completed by (initials): MDC

Discrepancy Documentation:

Any items above which are "No" or do not meet specifications (i.e. temps) must be resolved.

1. Person Contacted: _____ Phone Number: _____
   Initiated By: _____ Date: _____
   Problem:
   Resolution:
2. Person Contacted: _____ Phone Number: _____
   Initiated By: _____ Date: _____
   Problem:
   Resolution:

(Please use the back of this sheet for additional comments or cor

(4019696)
Sentry Ag Service
VI 2348406
iv 12/12/2023 08:25:15
VI 2348406

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Steven Meirinho

Name of Dairy Facility: Joe Meirinho and Son Dairy

Facility Address: 3354 W Whitmore Ave Modesto 95358
Number and Street City Zip Code

Contact Person Name and Phone Number: Steven Meirinho 209-538-8284
Name Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: Avila Spreading Service

Address of Hauling Company /Person: PO Box 58005
Number and Street City Zip Code

Contact Person: Lonny Avila 209-595-7955
Name Phone Number

Destination Information:

Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Number and Street	City	Zip Code	Phone Number
Nick Blom	2613 Illinois Ave	Modesto	95358	209-522-6140

Manure Destination Address or Assessor's Parcel Number:

Number and Street	City	Zip Code	Assessor's Parcel Number
2613 Illinois Ave	Modesto	95358	209-522-6140

Dates Hauled: 2/25/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 640 **Tons** ~~or Cubic Yards~~ (indicate which units used)
Manure Solids Content (if amount reported in tons): _____
Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: Load count / truck weight._____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: _____ Date: _____

Hauler's Signature: Tommy D. Curiel Date: _____